

My Explorations in the Collatz-Forest and Some of My Findings

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Some Ice-Breaker Pleasantries

A Caveat

A Disclaimer

we may not be in an integral domain;

but, i can assure you, there are

no irrationals; no radicals; no trans; no imaginaries; nothing complex;

not even any negatives;

only positives, that too, with some modular ideals.

HISTORICAL BACKGROUND

Lothar **C**ollatz (1910-1990);
Helmut **H**asse (1898-1979);

Bryan **T**hwaites (1923-);
Syracuse University, NY;

Stanislaw **U**lam (1909-1984);
Shizuo **K**akutani (1911-2004);

HISTORICAL BAGGAGE

Jeffrey C. Lagarias refers to the statement by **Paul Erdos**, one of the most revered Mathematician and Number Theory Expert, who said that - **"Mathematics is not yet ready for such problems"** – **"Hopeless, Absolutely Hopeless"** – about the Collatz Conjecture.

Richard K. Guy has been somewhat pessimistic in advising researchers - **"Don't Try to Solve These Problems"**.

Jeffrey C. Lagarias gives an elaborate overview of the Collatz Problem, in his book **"The Ultimate Challenge: The $3x+1$ Problem"**; and also an extensive two-part annotated bibliography.

"It is hard to resist exploring its structure. We should not exclude it from the mathematical universe just because we are unhappy with its difficulty. It is a fascinating and addictive problem".

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**The Collatz ( $3x+1$ ) System is indeed a highly 'chaotic' system defined or generated with one of the simplest deterministic well-defined equations, that too, just two in number.**

**Collatz Function****Collatz  $(3x+1)$  Sequence****Collatz Conjecture**

Start with any given positive integer  $x$ . If  $x$  is even then, divide by 2  $\{x \Leftarrow x/2\}$ ; else(if  $x$  is odd) then, multiply by 3 and add 1  $\{x \Leftarrow (3x+1)\}$  so as to get a resultant that is even. Now continue the process.

My way of defining the **Collatz Function**:

if  $(x \equiv 0 \pmod{2})$  then  $S(x) := [x]D := (x/2)$ ;  
 elseif  $(x \equiv 1 \pmod{2})$  then  $S(x) := [x]U := (3x+1)$ ;

The sequence thus generated, starting from any given positive integer, is the **Collatz Sequence** corresponding to that given positive integer.

**The Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Sequence**

The **Collatz Conjecture** states that the Collatz Sequence will eventually **converge** to the trivial-cycle; that is,  $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ ; starting from any positive integer.

## Collatz Sequence starting from 27

27, 82, 41, 124, 62, 31, 94, 47, 142, 71, 214, 107, 322, 161, 484, 242, 121, 364, 182, 91, 274, 137, 412, 206, 103, 310, 155, 466, 233, 700, 350, 175, 526, 263, 790, 395, 1186, 593, 1780, 890, 445, 1336, 668, 334, 167, 502, 251, 754, 377, 1132, 566, 283, 850, 425, 1276, 638, 319, 958, 479, 1438, 719, 2158, 1079, 3238, 1619, 4858, 2429, 7288, 3644, 1822, 911, 2734, 1367, 4102, 2051, 6154, 3077, 9232, 4616, 2308, 1154, 577, 1732, 866, 433, 1300, 650, 325, 976, 488, 244, 122, 61, 184, 92, 46, 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1

27, 82, 41, 124, 31, 94, 47, 142, 71, 214, 107, 322, 161, 484, 121, 364, 91, 274, 137, 412, 103, 310, 155, 466, 233, 700, 175, 526, 263, 790, 395, 1186, 593, 1780, 445, 1336, 167, 502, 251, 754, 377, 1132, 283, 850, 425, 1276, 319, 958, 479, 1438, 719, 2158, 1079, 3238, 1619, 4858, 2429, 7288, 911, 2734, 1367, 4102, 2051, 6154, 3077, 9232, 577, 1732, 866, 433, 1300, 650, 325, 976, 488, 244, 122, 61, 184, 92, 46, 23, 70, 35, 106, 53, 160, 5, 16, 1

27; 41; 31; 47; 71; 107; 161; 121; 91; 137; 103; 155; 233;  
175; 263; 395; 593; 445; 167; 251; 377; 283; 425;  
319; 479; 719; 1079; 1619; 2429; 911; 1367; 2051; 3077; 577; 433; 325; 61; 23;  
35; 53; 5; 1

CTUHSK-Generative-Parameter-Tuple for 27 : Reverse-Collatz-Halvings-Tuple

(4, 5, 1, 1, 3, 4, 2, 2, 4, 1, 1, 1, 3, 1, 1, 1, 1, 1, 2, 1, 2, 1, 1, 3, 2, 1, 1, 1, 2, 1, 1, 2, 1, 2, 2, 1, 1, 1, 1, 2, 1, 0)

## Compact Collatz (**even-to-odd, odd-to-even**) Function

if  $x \equiv 0 \pmod{2}$  then,  $T(x) := [x]D^\# := [x]D^* := [x]D^p = (x/2^p)$ ; for some valid  $p \geq 1$ ;

so as to get a resultant that is odd,  $x \not\equiv 0 \pmod{2}$ ;

elseIf  $x \not\equiv 0 \pmod{2}$  then,  $T(x) := [x]U^\# := [x]U := (3x+1)$ ; which is even,  $x \equiv 0 \pmod{2}$ .

The compact Collatz sequence is an alternating sequence of odd and even numbers.

## Condensed Compact (**odd-to-odd**) CTUHSK Function

Start with any positive odd integer  $x > 1$ ;  $x \not\equiv 0 \pmod{2}$ ; apply the operator ( $U^\#D^\#$ );

$x \leftarrow [x](U^\#D^\#)$ ; so that the resultant is odd,  $x \not\equiv 0 \pmod{2}$ ; now continue the process.

## Condensed Compact (**odd-to-odd**) Inverse CTUHSK Function

if  $\{(y \equiv 1 \pmod{6}) \vee (y \equiv 5 \pmod{6})\}$  then  $\{x\} := \{P(y)\} := (y \cdot 2^k - 1)/3$ ; to get a positive integer output  $x$ .

input  $y := 6b + x_6$ ; with  $0 \leq b$ ;  $b_3 := b \pmod{3} \in \{1, 2, 0\}$ ;  $y_3 := y \pmod{3} \in \{1, 2\}$  and  $y_6 := y \pmod{6} \in \{1, 5\}$ ;

binary-exponent-value  $k := 2j - (y_6 - y_3)/3$ ; for the  $j^{\text{th}}$  jump;  $1 \leq j$ ;  $j_3 := j \pmod{3} \in \{1, 2, 0\}$ ;

output  $x := 6a + x_6$ ; with  $0 \leq a$ ;  $a_3 := a \pmod{3} \in \{1, 2, 0\}$ ;  $x_3 := x \pmod{3} \in \{1, 2, 0\}$  and  $x_6 := x \pmod{6} \in \{1, 5, 3\}$ ;

## Modular Periodicity

| $[b]_3, [y]_6 =$       | (0,1)      | (0,5)       | (1,1)       | (1,5)       | (2,1)        | (2,5)       | $= [b]_3, [y]_6$       |
|------------------------|------------|-------------|-------------|-------------|--------------|-------------|------------------------|
| j=9; w=18<br>j=9; v=17 | (1,3)87381 | 218453(0,5) | (1,5)611669 | 480597(2,3) | (2,1)1135957 | 742741(1,1) | j=9; w=18<br>j=9; v=17 |
| j=8; w=16<br>j=8; v=15 | (1,5)21845 | 54613(0,1)  | (1,1)152917 | 120149(2,5) | (0,3)283989  | 185685(2,3) | j=8; w=16<br>j=8; v=15 |
| j=7; w=14<br>j=7; v=13 | (1,1)5461  | 13653(1,3)  | (2,3)38229  | 30037(2,1)  | (0,5)70997   | 46421(2,5)  | j=7; w=14<br>j=7; v=13 |
| j=6; w=12<br>j=6; v=11 | (2,3)1365  | 3413(1,5)   | (2,5)9557   | 1251(0,3)   | (0,1)17749   | 11605(2,1)  | j=6; w=12<br>j=6; v=11 |
| j=5; w=10<br>j=5; v=9  | (2,5)341   | 853(1,1)    | (2,1)2389   | 1877(0,5)   | (1,3)4437    | 2901(0,3)   | j=5; w=10<br>j=5; v=9  |
| j=4; w=8<br>j=4; v=7   | (2,1)85    | 213(2,3)    | (0,3)597    | 469(0,1)    | (1,5)1109    | 725(0,5)    | j=4; w=8<br>j=4; v=7   |
| j=3; w=6<br>j=3; v=5   | (0,3)21    | 53(2,5)     | (0,5)149    | 117(1,3)    | (1,1)277     | 181(0,1)    | j=3; w=6<br>j=3; v=5   |
| j=2; w=4<br>j=2; v=3   | (0,5)5     | 13(2,1)     | (0,1)37     | 29(1,5)     | (2,3)69      | 45(1,3)     | j=2; w=4<br>j=2; v=3   |
| j=1; w=2<br>j=1; v=1   | (0,1)1     | 3(0,3)      | (1,3)9      | 7(1,1)      | (2,5)17      | 11(1,5)     | j=1; w=2<br>j=1; v=1   |
| $[b]_3, [y]_6 =$       | (0,1)      | (0,5)       | (1,1)       | (1,5)       | (2,1)        | (2,5)       | $= [b]_3, [y]_6$       |

| $[b]_3, [y]_6 =$     | (0,1) | (0,5) | (1,1) | (1,5) | (2,1) | (2,5) | (0,1) | (0,5) | (1,1) | (1,5) | (2,1) | (2,5) | (0,1) | (0,5) | (1,1) | (1,5) | (2,1) | (2,5) | $= [b]_3, [y]_6$     |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------------------|
| j=1; w=2<br>j=1; v=1 | (0,1) | 3     | (1,3) | 7     | (2,5) | 11    | (1,1) | 15    | (2,3) | 19    | (0,5) | 23    | (2,1) | 27    | (0,3) | 31    | (1,5) | 35    | j=1; w=2<br>j=1; v=1 |
|                      | 1     | (0,3) | 9     | (1,1) | 17    | (1,5) | 25    | (2,3) | 33    | (0,1) | 41    | (0,5) | 49    | (1,3) | 57    | (2,1) | 65    | (2,5) |                      |
| $[b]_3, [y]_6 =$     | (0,1) | (0,5) | (1,1) | (1,5) | (2,1) | (2,5) | (0,1) | (0,5) | (1,1) | (1,5) | (2,1) | (2,5) | (0,1) | (0,5) | (1,1) | (1,5) | (2,1) | (2,5) | $= [b]_3, [y]_6$     |

## **Existence/Non-Existence of nontrivial-cycles/closed-loops in the Collatz System**

**if there exists a nontrivial-cycle/closed-loop in the Collatz system; then**

**its lowest odd-numbered-node must necessarily be of the form  $(4m-1)$**

**and cannot be of the form  $(4m+1)$ ;**

**although it may as well be of the form  $(6m+1)$  and possibly even of the form  $(6m-1)$ ;**

**its highest odd-numbered-node must necessarily be of the form  $(6m-1)$**

**and cannot be of the form  $(4m-1)$ ;**

**although it may as well be of the form  $(4m+1)$  and possibly even of the form  $(6m+1)$ ;**

## **Fruits of the CTUHSK Arborescence**

**Odd multiples of 3 reached through condensed compact inverse Collatz function applied on**

**$BEL(6b+1); 0 \leq b; (j_3-b_3)=3c;$**

**hanging from the rung defined by the binary-exponent-value  $k:=(2j):=2(3c+b_3); 0 \leq i;$**

**and**

**$BEL(6b+5); 0 \leq b; (j_3+b_3-1)=3c;$**

**hanging from the rung defined by the binary-exponent-value  $k:=(2j-1):=1+2(3c-b_3); 0 \leq i;$**

## SOME SIMPLER CREATURES

### Halemane's $(1.x+1)/2$ System (**even-to-odd, odd-to-even**) Form

Start with any given positive integer  $x$ ;

If  $x \equiv 0 \pmod{2}$  then, divide by 2 repeatedly, until the resultant  $x \equiv 1 \pmod{2}$ ;

ElseIf  $x \equiv 1 \pmod{2}$  then, add 1 to get a resultant  $x \equiv 0 \pmod{2}$ ; Now continue the process.

if  $x \equiv 0 \pmod{2}$  then,  $T(x) := [x]D^\# := [x]D^* := [x]D^p = (x/2^p)$ ;

elseif  $x \equiv 1 \pmod{2}$  then,  $T(x) := (1.x+1)$ ; so that the resultant is even.

### Inverse of $(1.x+1)/2$ System - Condensed Compact (**odd-to-odd**) Form

if  $y \equiv 1 \pmod{2}$  then  $\{P(y)\} := \{P_1(y, u) := (y \cdot 2^u - 1)\}; 1 \leq u$ .

OR

$y_0 := 1$ ; if  $y_k \equiv 1 \pmod{2}$  then  $y_{k+1} := y_k \cdot 2^{P_k} - 1$ ; else  $y_{k+1} := y_k \cdot 2^{P_k}$ ;

### Halemane's $(2.x \pm 1)/3$ System

Start with any given positive integer  $x$ ;

If  $x \equiv 0 \pmod{3}$  then, divide by 3 repeatedly until the resultant  $x \not\equiv 0 \pmod{3}$ ;

ElseIf  $x \equiv 1 \pmod{3}$  then, multiply by 2 and add 1 to get a resultant  $x \equiv 0 \pmod{3}$ ;

ElseIf  $x \equiv 2 \pmod{3}$  then, multiply by 2 and subtract 1 to get a resultant  $x \equiv 0 \pmod{3}$ ; Now continue the process.

### Inverse of $(2.x \pm 1)/3$ System

if  $\{(y \equiv 1 \pmod{6}) \vee ((y \equiv 5 \pmod{6}))\}$  then  $\{x\} := \{P(y)\} := (y \cdot 3^u \pm 1)/2$ ; with  $1 \leq u$ .

OR

$y_0 := 1$ ; if  $\{(y_k \equiv 1 \pmod{6}) \vee ((y_k \equiv 5 \pmod{6}))\}$  then  $y_{k+1} := (y_k \cdot 3^{P_k} \pm 1)/2$ ; else  $y_{k+1} := y_k \cdot 3^{P_k}$ ;

## Halemane's $(3.x \pm 1)/4$ System

Start with any given positive integer  $x > 1$ ; if  $(x=1)$  STOP;  
 elseif  $x \equiv 0 \pmod{4}$  then, divide by 4 repeatedly until the resultant  $x \not\equiv 0 \pmod{4}$ ;  
 ElseIf  $x \equiv 2 \pmod{4}$  then, divide by 2 to get a resultant  $x \not\equiv 2 \pmod{4}$ ;  
 ElseIf  $\{x \equiv 1 \pmod{4}\}$  then, multiply by 3 and add 1 to get a resultant  $\{x \equiv 0 \pmod{4}\}$ ;  
 ElseIf  $\{x \equiv 3 \pmod{4}\}$  then, multiply by 3 and subtract 1 to get a resultant  $\{x \equiv 0 \pmod{4}\}$ ;  
 Now continue the process.

## Halemane's $(3.x \pm 1)/4$ Function

if  $(x=1)$  STOP;  
 elseif  $x \equiv 0 \pmod{4}$  then,  $S(x) := (x/4)$ ;  
 elseif  $x \equiv 2 \pmod{4}$  then,  $S(x) := (x/2)$ ;  
 elseif  $\{x \equiv 1 \pmod{4}\}$  then,  $S(x) := (3.x+1)$ ;  
 elseif  $\{x \equiv 3 \pmod{4}\}$  then,  $S(x) := (3.x-1)$ ;

|         | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
|---------|---------|---------|---------|---------|
| $[0]_4$ | 0       | 1       | 1       | 1       |
| $[1]_4$ | 1       | 0       | 0       | 0       |
| $[2]_4$ | 0       | 1       | 0       | 1       |
| $[3]_4$ | 1       | 0       | 0       | 0       |

## condensed compact (odd-to-odd) inverse of Halemane's $(3.x \pm 1)/4$ function

if  $\{(y \equiv 1 \pmod{6}) \vee (y \equiv 5 \pmod{6})\}$  then  $\{x\} := \{P(y)\} := [4.y.2^u \pm 1]/3$ ; with  $0 \leq u$ ;  
 $P(y,u) := [y.2^u - (-1)^{(u^2 + (y^6 - y^3)/3)}]/3$  where  $u_2 := u \bmod 2$ ;  $y_6 := y \bmod 6$ ;  $y_3 := y \bmod 3$ .

### Halemane's $(3x \pm 1)/4$ System -vis-à-vis- Collatz $(3x+1)/2$ System

|    | 1               | 5               | 7          | 11          | 13          | 17          | 19          | 23          | 25          | 29          | 31          | 35          | 37          | 41          | 43          | 47          | 49          | 53          |    |
|----|-----------------|-----------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|----|
| 19 | 174763<br>(0,1) | 873813<br>(0,3) | 1223339    | 1922389     | 2271915     | 2970965     | 3320491     | 4019541     | 4369067     | 5068117     | 5417643     | 6116693     | 6466219     | 7165269     | 7514795     | 8213845     | 8563371     | 9262421     | 19 |
| 18 | 87381<br>(1,3)  | 436907<br>(1,5) | 611669     | 961195      | 1135957     | 1485483     | 1660245     | 2009771     | 2184533     | 2534059     | 2708821     | 3058347     | 3233109     | 3582635     | 3757397     | 4106923     | 4281685     | 4631211     | 18 |
| 17 | 43691<br>(0,5)  | 218453<br>(0,5) | 305835     | 480597      | 567979      | 742741      | 830123      | 1004885     | 1092267     | 1267029     | 1354411     | 1529173     | 1616555     | 1791317     | 1878699     | 2053461     | 2140843     | 2315605     | 17 |
| 16 | 21845<br>(1,5)  | 109227<br>(0,3) | 152917     | 240299      | 283989      | 371371      | 415061      | 502443      | 546133      | 633515      | 677205      | 764587      | 808277      | 895659      | 939349      | 1026731     | 1070421     | 1157803     | 16 |
| 15 | 10923<br>(2,3)  | 54613<br>(0,1)  | 76459      | 120149      | 141995      | 185685      | 207531      | 251221      | 273067      | 316757      | 338603      | 382293      | 404139      | 447829      | 469675      | 513365      | 535211      | 578901      | 15 |
| 14 | 5461<br>(1,1)   | 27307<br>(0,1)  | 38229      | 60075       | 70997       | 92843       | 103765      | 125611      | 136533      | 158379      | 169301      | 191147      | 202069      | 223915      | 234837      | 256683      | 267605      | 289451      | 14 |
| 13 | 2731<br>(2,1)   | 13653<br>(1,3)  | 19115      | 30037       | 35499       | 46421       | 51883       | 62805       | 68267       | 79189       | 84651       | 95573       | 101035      | 111957      | 117419      | 128341      | 133803      | 144725      | 13 |
| 12 | 1365<br>(2,3)   | 6827<br>(0,5)   | 9557       | 15019       | 17749       | 23211       | 25941       | 31403       | 34133       | 39595       | 42325       | 47787       | 50517       | 55979       | 58709       | 64171       | 66901       | 72363       | 12 |
| 11 | 683<br>(2,5)    | 3413<br>(1,5)   | 4779       | 7509        | 8875        | 11605       | 12971       | 15701       | 17067       | 19797       | 21163       | 23893       | 25259       | 27989       | 29355       | 32085       | 33451       | 36181       | 11 |
| 10 | 341<br>(2,5)    | 1707<br>(2,3)   | 2389       | 3755        | 4437        | 5803        | 6485        | 7851        | 8533        | 9899        | 10581       | 11947       | 12629       | 13995       | 14677       | 16043       | 16725       | 18091       | 10 |
| 9  | 171<br>(1,3)    | 853<br>(1,1)    | 1195       | 1877        | 2219        | 2901        | 3243        | 3925        | 4267        | 4949        | 5291        | 5973        | 6315        | 6997        | 7339        | 8021        | 8363        | 9045        | 9  |
| 8  | 85<br>(2,1)     | 427<br>(2,1)    | 597        | 939         | 1109        | 1451        | 1621        | 1963        | 2133        | 2475        | 2645        | 2987        | 3157        | 3499        | 3669        | 4011        | 4181        | 4523        | 8  |
| 7  | 43<br>(1,1)     | 213<br>(2,3)    | 299        | 469         | 555         | 725         | 811         | 981         | 1067        | 1237        | 1323        | 1493        | 1579        | 1749        | 1835        | 2005        | 2091        | 2261        | 7  |
| 6  | 21<br>(0,3)     | 107<br>(2,5)    | 149        | 235         | 277         | 363         | 405         | 491         | 533         | 619         | 661         | 747         | 789         | 875         | 917         | 1003        | 1045        | 1131        | 6  |
| 5  | 11<br>(1,5)     | 53<br>(2,5)     | 75         | 117         | 139         | 181         | 203         | 245         | 267         | 309         | 331         | 373         | 395         | 437         | 459         | 501         | 523         | 565         | 5  |
| 4  | 5<br>(0,5)      | 27<br>(1,3)     | 37         | 59          | 69          | 91          | 101         | 123         | 133         | 155         | 165         | 187         | 197         | 219         | 229         | 251         | 261         | 283         | 4  |
| 3  | 3<br>(0,3)      | 13<br>(2,1)     | 19         | 29          | 35          | 45          | 51          | 61          | 67          | 77          | 83          | 93          | 99          | 109         | 115         | 125         | 131         | 141         | 3  |
| 2  | 1<br>(0,1)      | 7<br>(1,1)      | 9<br>(1,3) | 15<br>(2,3) | 17<br>(2,5) | 23<br>(0,5) | 25<br>(1,1) | 31<br>(2,1) | 33<br>(2,3) | 39<br>(0,3) | 41<br>(0,5) | 47<br>(1,5) | 49<br>(2,1) | 55<br>(0,1) | 57<br>(0,3) | 63<br>(1,3) | 65<br>(1,5) | 71<br>(2,5) | 2  |
| 1  |                 | 3               |            | 7           |             | 11          |             | 15          |             | 19          |             | 23          |             | 27          |             | 31          |             | 35          | 1  |
|    | 1               | 5               | 7          | 11          | 13          | 17          | 19          | 23          | 25          | 29          | 31          | 35          | 37          | 41          | 43          | 47          | 49          | 53          |    |

### Collatz (3x+1)/2 System -vis-à-vis- Halemene's (3.x±1)/4 System

|    | 1               | 5               | 7          | 11          | 13          | 17          | 19          | 23          | 25          | 29          | 31          | 35          | 37          | 41          | 43          | 47          | 49          | 53          |    |
|----|-----------------|-----------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|----|
| 19 | 174763<br>(0,1) | 873813<br>(0,3) | 1223339    | 1922389     | 2271915     | 2970965     | 3320491     | 4019541     | 4369067     | 5068117     | 5417643     | 6116693     | 6466219     | 7165269     | 7514795     | 8213845     | 8563371     | 9262421     | 19 |
| 18 | 87381<br>(1,3)  | 436907<br>(1,5) | 611669     | 961195      | 1135957     | 1485483     | 1660245     | 2009771     | 2184533     | 2534059     | 2708821     | 3058347     | 3233109     | 3582635     | 3757397     | 4106923     | 4281685     | 4631211     | 18 |
| 17 | 43691<br>(0,5)  | 218453<br>(0,5) | 305835     | 480597      | 567979      | 742741      | 830123      | 1004885     | 1092267     | 1267029     | 1354411     | 1529173     | 1616555     | 1791317     | 1878699     | 2053461     | 2140843     | 2315605     | 17 |
| 16 | 21845<br>(1,5)  | 109227<br>(0,3) | 152917     | 240299      | 283989      | 371371      | 415061      | 502443      | 546133      | 633515      | 677205      | 764587      | 808277      | 895659      | 939349      | 1026731     | 1070421     | 1157803     | 16 |
| 15 | 10923<br>(2,3)  | 54613<br>(0,1)  | 76459      | 120149      | 141995      | 185685      | 207531      | 251221      | 273067      | 316757      | 338603      | 382293      | 404139      | 447829      | 469675      | 513365      | 535211      | 578901      | 15 |
| 14 | 5461<br>(1,1)   | 27307<br>(0,1)  | 38229      | 60075       | 70997       | 92843       | 103765      | 125611      | 136533      | 158379      | 169301      | 191147      | 202069      | 223915      | 234837      | 256683      | 267605      | 289451      | 14 |
| 13 | 2731<br>(2,1)   | 13653<br>(1,3)  | 19115      | 30037       | 35499       | 46421       | 51883       | 62805       | 68267       | 79189       | 84651       | 95573       | 101035      | 111957      | 117419      | 128341      | 133803      | 144725      | 13 |
| 12 | 1365<br>(2,3)   | 6827<br>(0,5)   | 9557       | 15019       | 17749       | 23211       | 25941       | 31403       | 34133       | 39595       | 42325       | 47787       | 50517       | 55979       | 58709       | 64171       | 66901       | 72363       | 12 |
| 11 | 683<br>(2,5)    | 3413<br>(1,5)   | 4779       | 7509        | 8875        | 11605       | 12971       | 15701       | 17067       | 19797       | 21163       | 23893       | 25259       | 27989       | 29355       | 32085       | 33451       | 36181       | 11 |
| 10 | 341<br>(2,5)    | 1707<br>(2,3)   | 2389       | 3755        | 4437        | 5803        | 6485        | 7851        | 8533        | 9899        | 10581       | 11947       | 12629       | 13995       | 14677       | 16043       | 16725       | 18091       | 10 |
| 9  | 171<br>(1,3)    | 853<br>(1,1)    | 1195       | 1877        | 2219        | 2901        | 3243        | 3925        | 4267        | 4949        | 5291        | 5973        | 6315        | 6997        | 7339        | 8021        | 8363        | 9045        | 9  |
| 8  | 85<br>(2,1)     | 427<br>(2,1)    | 597        | 939         | 1109        | 1451        | 1621        | 1963        | 2133        | 2475        | 2645        | 2987        | 3157        | 3499        | 3669        | 4011        | 4181        | 4523        | 8  |
| 7  | 43<br>(1,1)     | 213<br>(2,3)    | 299        | 469         | 555         | 725         | 811         | 981         | 1067        | 1237        | 1323        | 1493        | 1579        | 1749        | 1835        | 2005        | 2091        | 2261        | 7  |
| 6  | 21<br>(0,3)     | 107<br>(2,5)    | 149        | 235         | 277         | 363         | 405         | 491         | 533         | 619         | 661         | 747         | 789         | 875         | 917         | 1003        | 1045        | 1131        | 6  |
| 5  | 11<br>(1,5)     | 53<br>(2,5)     | 75         | 117         | 139         | 181         | 203         | 245         | 267         | 309         | 331         | 373         | 395         | 437         | 459         | 501         | 523         | 565         | 5  |
| 4  | 5<br>(0,5)      | 27<br>(1,3)     | 37         | 59          | 69          | 91          | 101         | 123         | 133         | 155         | 165         | 187         | 197         | 219         | 229         | 251         | 261         | 283         | 4  |
| 3  | 3<br>(0,3)      | 13<br>(2,1)     | 19         | 29          | 35          | 45          | 51          | 61          | 67          | 77          | 83          | 93          | 99          | 109         | 115         | 125         | 131         | 141         | 3  |
| 2  | 1<br>(0,1)      | 7<br>(1,1)      | 9<br>(1,3) | 15<br>(2,3) | 17<br>(2,5) | 23<br>(0,5) | 25<br>(1,1) | 31<br>(2,1) | 33<br>(2,3) | 39<br>(0,3) | 41<br>(0,5) | 47<br>(1,5) | 49<br>(2,1) | 55<br>(0,1) | 57<br>(0,3) | 63<br>(1,3) | 65<br>(1,5) | 71<br>(2,5) | 2  |
| 1  |                 | 3               |            | 7           |             | 11          |             | 15          |             | 19          |             | 23          |             | 27          |             | 31          |             | 35          | 1  |
|    | 1               | 5               | 7          | 11          | 13          | 17          | 19          | 23          | 25          | 29          | 31          | 35          | 37          | 41          | 43          | 47          | 49          | 53          |    |

## Halemane's $(3.x+1)/4$ System - as a **clone** of the Collatz System

Start with any given positive integer  $x > 1$ ; if  $(x=1)$  STOP;  
 elseif  $x \equiv 0 \pmod{4}$  then, divide by 4 repeatedly until the resultant  $x \not\equiv 0 \pmod{4}$ ;  
 ElseIf  $x \equiv 2 \pmod{4}$  then, **multiply** by 2 to get a resultant  $x \equiv 0 \pmod{4}$ ;  
 ElseIf  $\{x \equiv 1 \pmod{4}\} \vee \{x \equiv 3 \pmod{4}\}$  then, multiply by 3 and add 1 to get a resultant  $\{x \equiv 0 \pmod{4}\} \vee \{x \equiv 2 \pmod{4}\}$ ; Now continue the process.

## Halemane's $(3.x+1)/4$ Function

if  $(x=1)$  STOP;  
 elseif  $x \equiv 0 \pmod{4}$  then,  $S(x) := (x/4)$ ; elseif  $x \equiv 2 \pmod{4}$  then,  $S(x) := (2x)$ ;  
 elseif  $\{x \equiv 1 \pmod{4}\} \vee \{x \equiv 3 \pmod{4}\}$  then,  $S(x) := (3.x+1)$ ;

|         | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
|---------|---------|---------|---------|---------|
| $[0]_4$ | 0       | 1       | 1       | 1       |
| $[1]_4$ | 1       | 0       | 0       | 0       |
| $[2]_4$ | 1       | 0       | 0       | 0       |
| $[3]_4$ | 0       | 0       | 1       | 0       |

## Halemane's $(3.x+1)/4$ System - **modified**

Start with any given positive integer  $x > 1$ ; if  $(x=1)$  STOP;  
 elseif  $x \equiv 0 \pmod{4}$  then, divide by 4 repeatedly until the resultant  $x \not\equiv 0 \pmod{4}$ ;  
 ElseIf  $x \equiv 2 \pmod{4}$  then, **divide** by 2 to get a resultant  $x \not\equiv 2 \pmod{4}$ ;  
 ElseIf  $\{x \equiv 1 \pmod{4}\} \vee \{x \equiv 3 \pmod{4}\}$  then, multiply by 3 and add 1 to get a resultant  $\{x \equiv 0 \pmod{4}\} \vee \{x \equiv 2 \pmod{4}\}$ ; Now continue the process.

## Halemane's $(3.x+1)/4$ Function

if  $(x=1)$  STOP;  
 elseif  $x \equiv 0 \pmod{4}$  then,  $S(x) := (x/4)$ ; elseif  $x \equiv 2 \pmod{4}$  then,  $S(x) := (x/2)$ ;  
 elseif  $\{x \equiv 1 \pmod{4}\} \vee \{x \equiv 3 \pmod{4}\}$  then,  $S(x) := (3.x+1)$ ;

|         | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
|---------|---------|---------|---------|---------|
| $[0]_4$ | 0       | 1       | 1       | 1       |
| $[1]_4$ | 1       | 0       | 0       | 0       |
| $[2]_4$ | 0       | 1       | 0       | 1       |
| $[3]_4$ | 0       | 0       | 1       | 0       |

## Recommended Reading

- [§1]. Wikipedia Page – [https://en.wikipedia.org/wiki/Collatz\\_conjecture](https://en.wikipedia.org/wiki/Collatz_conjecture)
- [§2]. Richard K Guy; “Don’t Try to Solve These Problems”;  
The American Mathematical Monthly: Vol. 90, No. 1, pp. 35-41; 1983.  
<https://doi.org/10.1080/00029890.1983.11971148>
- [§3]. Jeffrey C Lagarias; “The  $3x+1$  problem: An annotated bibliography (1963--1999); sorted by author”;  
<https://arxiv.org/abs/math/0309224>
- [§4]. Jeffrey C Lagarias; “The  $3x+1$  Problem: An Annotated Bibliography, II (2000-2009)”;  
<https://arxiv.org/abs/math/0608208>
- [§5]. Jeffrey C Lagarias; “The  $3x + 1$  Problem: An Overview”;  
<https://arxiv.org/abs/2111.02635>
- [§6]. Jeffrey C Lagarias; “The Ultimate Challenge: The  $3x+1$  Problem”;  
American Mathematical Society; 2010.
- [§7]. Keshava Prasad Halemane; “Collatz-Conjecture Proved and Halemane-Conjecture Proposed”;  
<https://doi.org/10.31224/6063>; 2025.
- [§8]. Keshava Prasad Halemane; “Halemane’s Theory on the Collatz ( $3x+1$ ) System”;  
<https://doi.org/10.31224/7569>; 2026.

## EXPLORATORY-ROUGH-WORK

$$[6k-1] = [4k-1](UD);$$

if  $[m](UD)^k = n$ ; then  $(n+1)/(m+1) = (3/2)^k$ ;  
 or equivalently,  $(n+1).2^k = (m+1).3^k$ ;  
 or equivalently,  $[2.2^k - 1](UD)^k = [2.3^k - 1]$ ;

If  $x \equiv 0 \pmod{2}$ ; let  $x = (a.2^p - 1).2^q$  for some  $a \equiv 1 \pmod{2}$  and  $0 < p$  and  $0 < q$ ;  
 then,  $x \Leftarrow [x](D)^q = [x]/2^q = (a.2^p - 1)$ ; which is an odd number;  
 Elseif  $x \equiv 1 \pmod{2}$  and if  $x = (a.2.2^p - 1)$  for some  $a \equiv 1 \pmod{2}$  and  $0 \leq p$ ;  
 then,  $x \Leftarrow [x](UD)^q = (a.2.3^p - 1)$ ; which is an odd number.

$$[6k+1] = [8k+1](UDD) ;$$

if  $[m](UDD)^k = n$ ; then  $(n-1)/(m-1) = (3/4)^k$ ;  
 or equivalently,  $(n-1).4^k = (m-1).3^k$ ;  
 or equivalently,  $[4^k + 1](UDD)^k = [3^k + 1]$ ;

If  $x \equiv 0 \pmod{2}$ ; let  $x = (a.2^p - 1).4^q$  for some  $a \equiv 1 \pmod{2}$  and  $0 < p$  and  $0 < q$ ;  
 then,  $x \Leftarrow [x](DD)^q = [x]/4^q = (a.2^p - 1)$ ; which is an odd number;  
 Elseif  $x \equiv 1 \pmod{2}$  and if  $x = (a.4.4^q + 1)$  for some  $a \equiv 1 \pmod{2}$  and  $0 \leq q$ ;  
 then,  $x \Leftarrow [x](UDD)^q = (a.4.3^q - 1)$ ; which is an odd number.

$$z = (6b + 1) \Leftarrow (8b + 1) = y = (6a - 1) \Leftarrow (4a - 1) = x$$

$$4(z - 1) = 3(y - 1) \text{ and } 2(y + 1) = 3(x + 1)$$

$$8(z - 1) = 6(y - 1) \text{ and } 6(y + 1) = 9(x + 1)$$

$$2(4z - 1) = 6(y) = 3(3x + 1)$$

| <b>z</b>  | <b><math>z \Leftarrow y</math></b>                | <b>y</b>  | <b>!=!</b> | <b>y</b>  | <b><math>y \Leftarrow x</math></b>                   | <b>x</b>  |
|-----------|---------------------------------------------------|-----------|------------|-----------|------------------------------------------------------|-----------|
| <b>1</b>  | <b><math>z \Leftarrow y(\text{UDD})D^2</math></b> | <b>5</b>  |            | <b>5</b>  | <b><math>y \Leftarrow x(\text{UD})</math></b>        | <b>3</b>  |
| <b>5</b>  | <b><math>z \Leftarrow y(\text{UDD})D</math></b>   | <b>13</b> |            | <b>5</b>  | <b><math>y \Leftarrow x(\text{UD})^4(D)^3</math></b> | <b>15</b> |
| <b>13</b> | <b><math>z \Leftarrow y(\text{UDD})</math></b>    | <b>17</b> |            | <b>17</b> | <b><math>y \Leftarrow x(\text{UD})</math></b>        | <b>11</b> |
| <b>7</b>  | <b><math>z \Leftarrow y(\text{UDD})</math></b>    | <b>9</b>  |            | <b>11</b> | <b><math>y \Leftarrow x(\text{UD})</math></b>        | <b>7</b>  |
| <b>11</b> | <b><math>z \Leftarrow y(\text{UDD})D</math></b>   | <b>29</b> |            | <b>29</b> | <b><math>y \Leftarrow x(\text{UD})</math></b>        | <b>19</b> |
| <b>19</b> | <b><math>z \Leftarrow y(\text{UDD})</math></b>    | <b>25</b> |            | <b>35</b> | <b><math>y \Leftarrow x(\text{UD})</math></b>        | <b>23</b> |
| <b>25</b> | <b><math>z \Leftarrow y(\text{UDD})</math></b>    | <b>33</b> |            | <b>47</b> | <b><math>y \Leftarrow x(\text{UD})</math></b>        | <b>31</b> |
| <b>31</b> | <b><math>z \Leftarrow y(\text{UDD})</math></b>    | <b>41</b> |            | <b>41</b> | <b><math>y \Leftarrow x(\text{UD})</math></b>        | <b>27</b> |
| <b>z</b>  | <b><math>z \Leftarrow y</math></b>                | <b>y</b>  |            | <b>y</b>  | <b><math>y \Leftarrow x</math></b>                   | <b>x</b>  |

$$1 \Leftarrow 5(\text{UDD})^2; 5 \Leftarrow 13(\text{UDD})D; 13 \Leftarrow 17(\text{UDD}); 17 \Leftarrow 11(\text{UD}); 11 \Leftarrow 7(\text{UD}); 7 \Leftarrow 9(\text{UDD}); \\ 5 \Leftarrow 3(\text{UD}); 5 \Leftarrow 15(\text{UD})^4(D)^3; 1 \Leftarrow 21(\text{UDD})(D)^4;$$